

CSC-206 Software Engineering
Project Report

FF BANK NET BANKING

Group -19

Name of the members	Enrollment Number	Mobile Number
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Contributions:

V Jeevan Sooriya: Designed and implemented the frontend of users and handled the routes of the corresponding pages.

Jagannadham : Designed and implemented the front end of the admin.

Karthik :designed and implemented logic for backend .

Nikhil kumar : Database creation and uploading data in it,server.

Introduction

Background Topics

Banking systems are critical infrastructures that manage financial transactions, customer data, and services like loans, fund transfers, and cheques. As financial operations move online, the demand for secure, user-friendly, and efficient banking platforms has grown.

Motivation of the Project

The goal of this project is to design a simplified, full-stack banking application with features that mimic real-world banking services, such as User authentication (signup/login),Money transfers,Loan management with EMI payments,Cheque processing

This system can be extended to real-world microfinance institutions, educational projects, or internal bank prototypes.

Application Domain

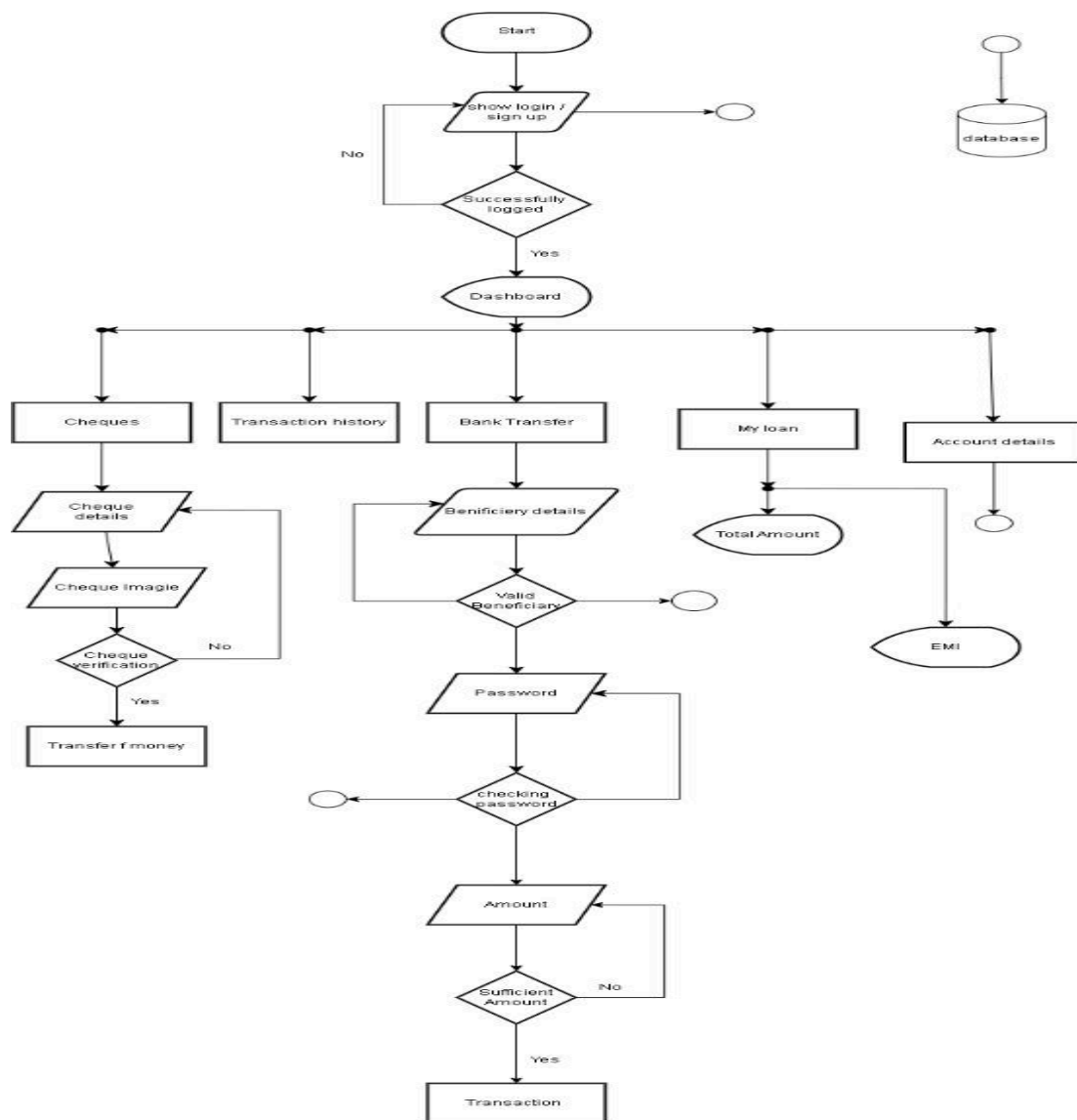
The project belongs to the **financial technology (fintech)** domain. It applies to Online banking systems, Loan management platforms, Cheque and EMI automation systems and Admin-user management systems

Proposed System

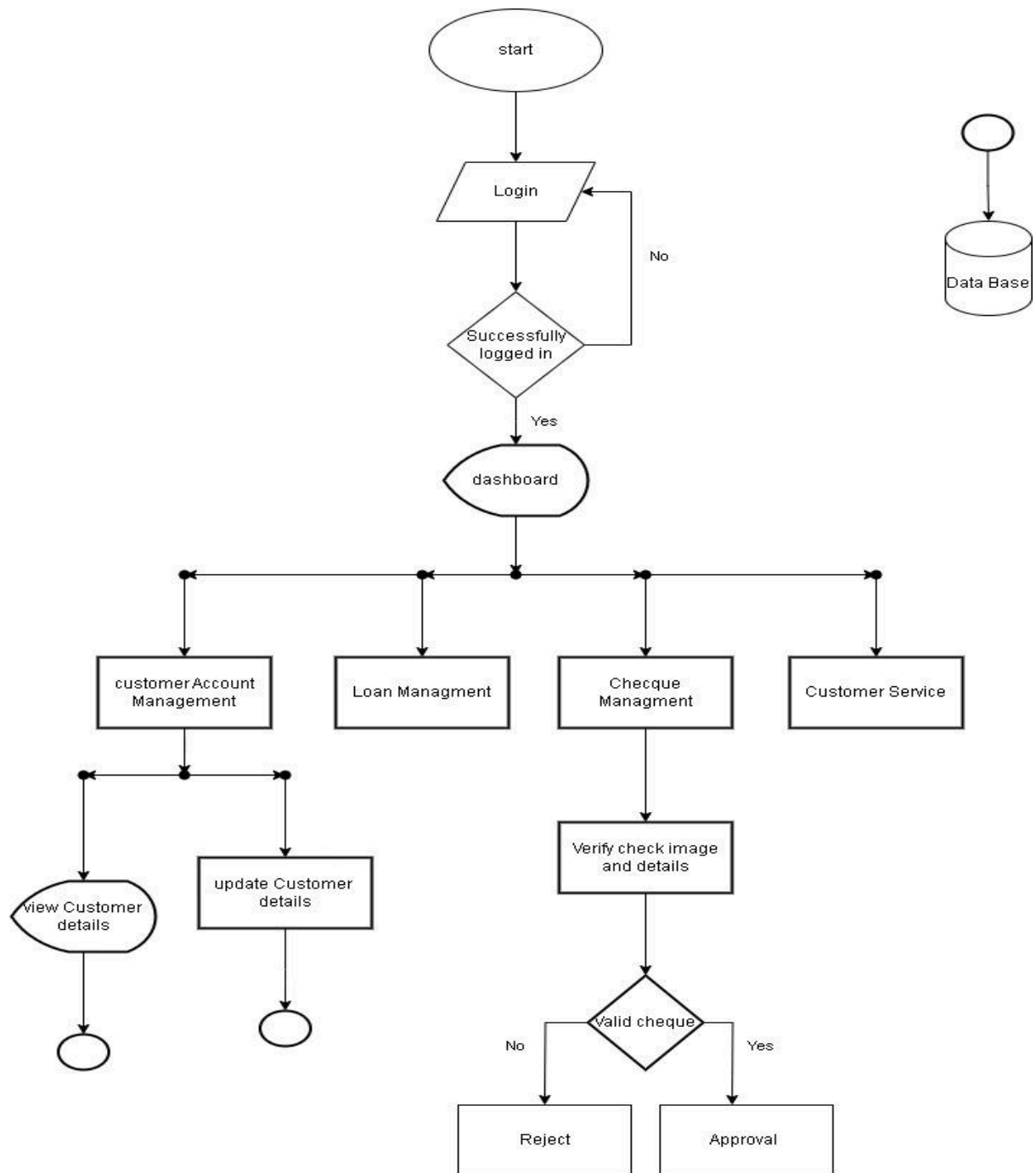
System Overview

The proposed system is a web-based application that provides both customer and admin interfaces. Key features are Secure login/signup with credentials, Fund transfer using account number and PIN, EMI calculation and payment, Cheque image upload and admin approval and Admin dashboard for user and cheque management

Application of Flowchart for User



Application of Flowchart for Admin



Steps to use the app:

The system consists of a sequence of interconnected modules that operate in the following steps:

1. User Signup/Login

- Users register using account number and password.
- Credentials are securely stored and verified on login.
- Authenticated users are redirected to the dashboard.

2. Account Overview

- The user dashboard displays current account balance and links to services like transfers, loans, and cheques.

3. Fund Transfer

- Users initiate a transfer by entering the recipient account number, amount, and password.
- The system authenticates the user, checks balance, updates sender/receiver balances, and logs the transaction.

4. Loan Management

- Users view active loans with details like amount, interest rate, EMI, and status. Loans are stored in the database with a status flag (pending, approved, repaid).
- Users select a loan, enter their password, and pay the calculated EMI if the loan is approved.
- The system checks balance, updates the loan status, deducts EMI, and logs the payment.

5. Cheque Upload and Processing

- Users upload a digital cheque image, fill in sender/receiver details, and submit.
- Admin reviews cheques, verifies authenticity, and approves/rejects.

6. Admin Interface

- Admins have a separate login. They can manage user accounts, view transactions, approve loans, and validate cheques.

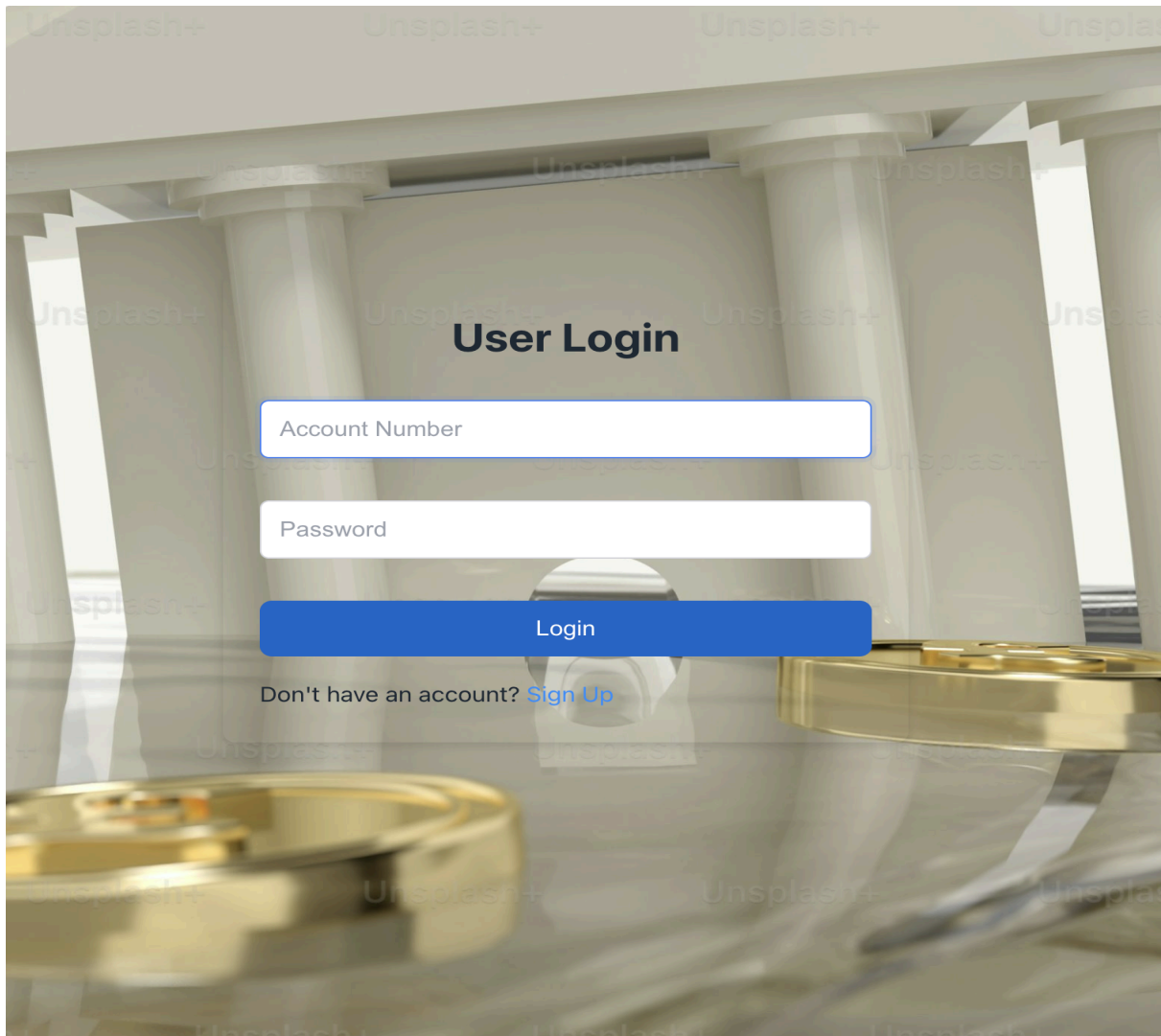
Implementation

Implementation Details

The system was implemented as a full-stack web application that simulates a core banking system. The implementation includes the following key modules

1. User Authentication Module

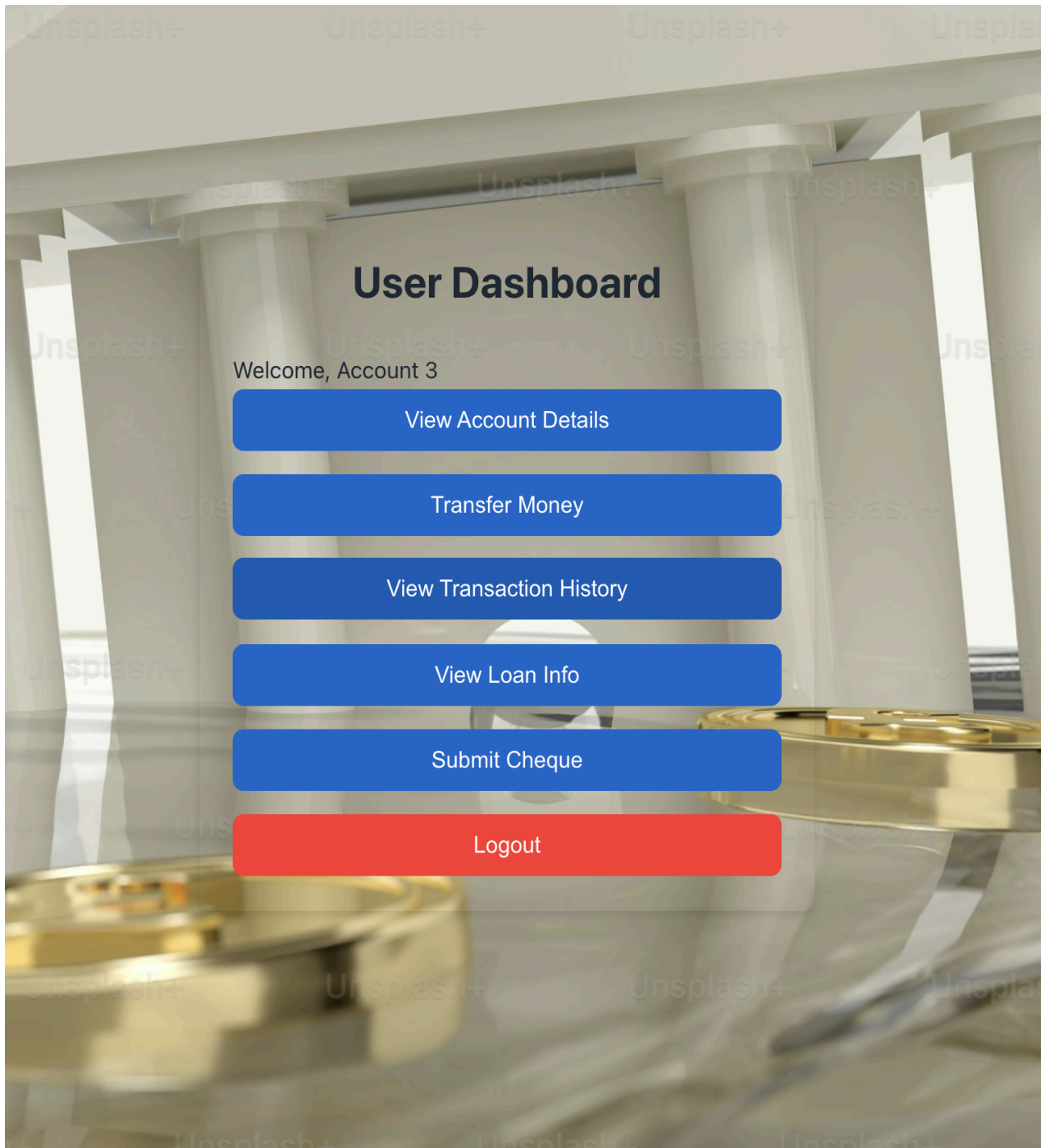
Signup and login functionality using account number and password.



2.Account Dashboard

Displays user details and available balance.

Provides navigation to fund transfer, loan management, and cheque upload modules.



3.Transaction Module

Users can transfer funds between accounts.

All transactions are recorded with type, description, and timestamp.

Back to Dashboard

Transfer successful

Recipient Account Number

Amount

PIN

Please fill in this field.

Description (optional)

Transfer

1. Loan Management Module

Users can view loan details and pay EMIs.

EMI payments are password-authenticated and deducted from the account balance.

Back to Dashboard

Loan ID	Amount	Interest Rate	Status	EMI	Created At	Select
1	₹50000	7.50%	approved	₹4338	22/04/2025, 17:49:00	

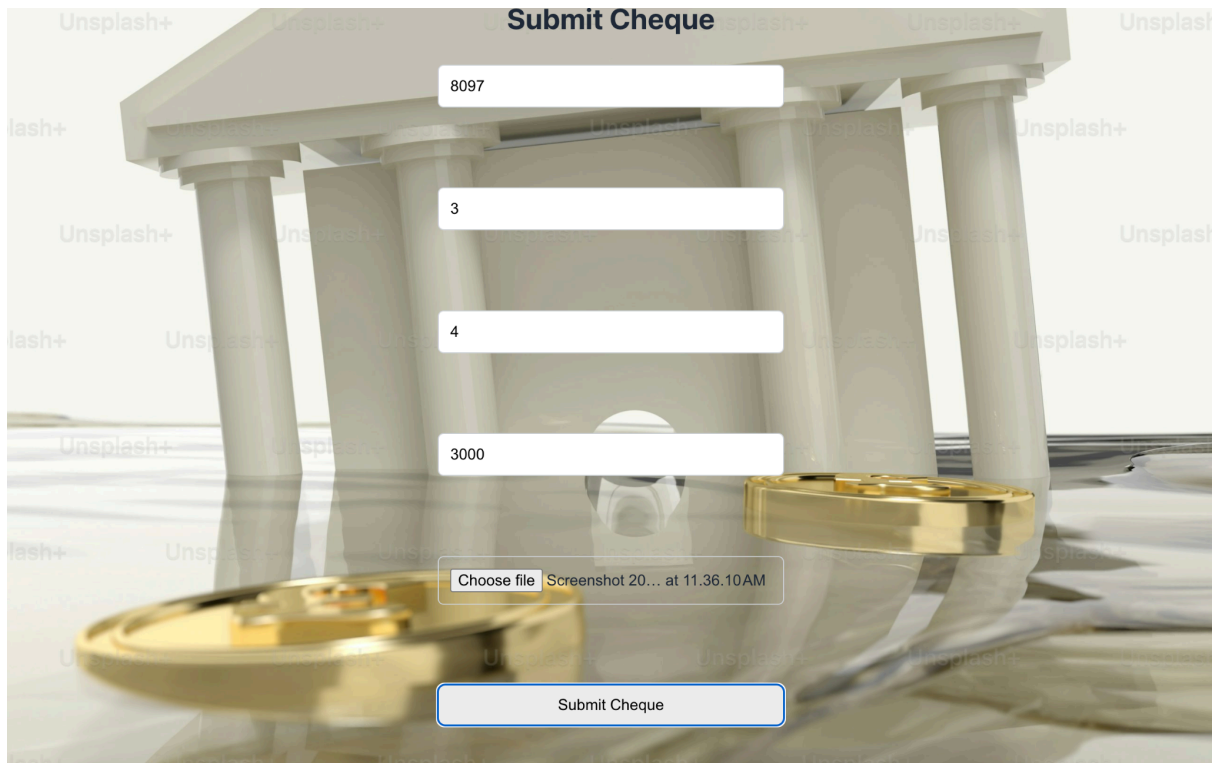
Pay EMI

Enter your password

Confirm EMI Payment

2. Cheque Handling Module

Users upload cheque images.



The 'Submit Cheque' form is overlaid on a background image of a classical building with columns and a large gold coin. The form contains five input fields: a text field with '8097', a text field with '3', a text field with '4', a text field with '3000', and a file upload field with the text 'Choose file' and 'Screenshot 20... at 11.36.10AM'. Below these fields is a blue 'Submit Cheque' button.

Submit Cheque

8097

3

4

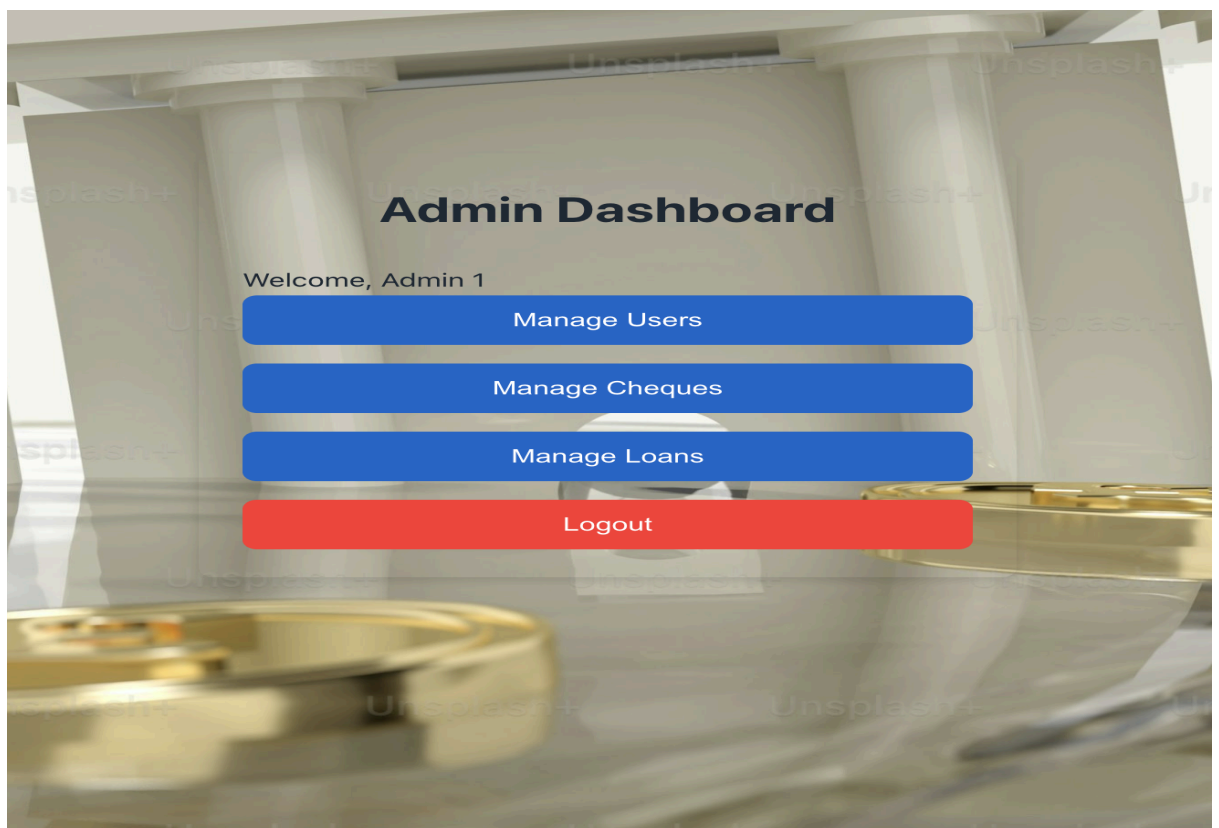
3000

Choose file Screenshot 20... at 11.36.10AM

Submit Cheque

3. Admin Panel

Admins can log in to view pending cheques, approve loans, and manage users.



The 'Admin Dashboard' is overlaid on the same background image as the 'Submit Cheque' form. It features a welcome message 'Welcome, Admin 1' and four buttons: 'Manage Users', 'Manage Cheques', 'Manage Loans', and 'Logout'. The first three buttons are blue, and the 'Logout' button is red.

Admin Dashboard

Welcome, Admin 1

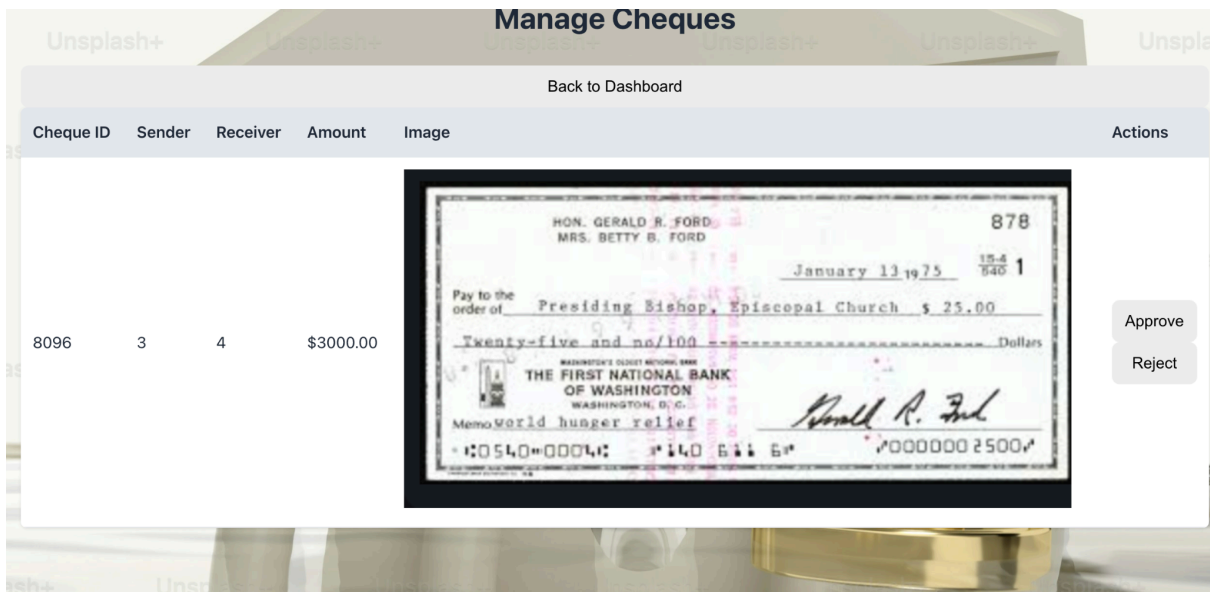
Manage Users

Manage Cheques

Manage Loans

Logout

Admin can view, approve, or reject cheques based on the image and details.



Feature	Our system	Usual bank portal
User Authentication	Account number and password	Username & OTP
Fund Transfers	Yes, with password authentication	Yes
EMI Payments	Password-verified EMI payment	Yes
Loan Tracking	Full loan list + status	Yes
Cheque Upload & Approval	Digital image upload + admin approval	Physical submission

Conclusion

This project successfully demonstrates the development of a full-featured digital banking system designed to replicate essential functionalities of a real-world banking portal. From user authentication and fund transfers to EMI payments, loan tracking, and cheque management, the system provides a streamlined and intuitive experience for both users and administrators.

Through modular architecture and the use of modern technologies like React.js and Node.js, the system ensures scalability, maintainability, and a responsive user interface. The integration with MySQL databases allows for efficient data management and secure transaction handling.

One of the key achievements of this project is the implementation of real-world features such as password-verified EMI payments and digital cheque uploads, offering a practical perspective on how banking software is developed and maintained. The admin dashboard brings in operational oversight, making the application viable for enterprise-level use with further extensions.

Overall, the project bridges theoretical concepts with real-world application, demonstrating both technical competency and an understanding of core banking workflows. Future improvements can include addition of otp and addition of fingerprint.

References:

<https://react.dev/>

<https://www.w3schools.com/html/default.asp>

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<https://www.w3schools.com/mysql/default.asp>

<https://www.w3schools.com/nodejs/default.asp>